Sukrit Arora

sukrit.arora@berkeley.edu · linkedin.com/in/sukritarora · San Francisco, CA · US Citizen

Experience

Deepnight (YC) — San Francisco, CA
Founding Research Engineer & Tech Lead — *Apr. 2024 – Present*

- Architected an end-to-end ML pipeline for real-time video denoising on embedded hardware: spanning raw sensor data collection, probabilistic noise modeling, model architecture design, multi-GPU training infrastructure, and production edge deployment.
- Achieved state-of-the-art low-light video performance: **90 FPS at 6.4 MP resolution at under 1W power** on Qualcomm and NVIDIA embedded SoCs. [Seeing Cats in the Dark]
- Developed quantization-aware training (QAT) pipelines targeting Qualcomm AI Engine (SNPE/QNN) and NVIDIA Jetson; INT8 models preserve perceptual quality on real-world night scenes.
- Led technical roadmap and sprint planning as Tech Lead; contributed to Series A fundraising through investor technical demos and due diligence.

KLA — Ann Arbor, MI
AI/ML Algorithm Engineer — *Aug. 2021 – Apr. 2024*

- Designed an unsupervised generative AI segmentation system for zero-annotation anomaly detection in SEM images, eliminating manual labeling and **reducing time-to-deployed-model by 4+ hours** per production run.
- Built a deep metric learning reverse image search system (contrastive/triplet loss) across multi-source defect datasets, improving classification model performance by **60%** through higher-quality training data curation.
- Implemented network saliency visualization to expose model reasoning for semiconductor and government clients, improving field debugging and customer confidence.
- Secured a competitive research grant for a KLA–UW-Madison collaboration on continual deep learning for image segmentation; managed partnership from proposal through execution.
- Mentored two summer interns; delivered internal deep learning course to 20+ engineers. Received Engineer of the Month and Team of the Month awards.

UC Berkeley EECS — Berkeley, CA
Graduate Student Researcher [Thesis] — Advised by Michael Lustig and Stella Yu — *Jan. 2020 – May 2021*

- Developed an image denoising and subsampled 3D MRI reconstruction system using a deep decoder generative network; published at ISMRM 2020 (cited by 32). [Conference Poster]
- Devised a technique for suppressing overlapping artifacts in volumetric MR image stitching.

Undergraduate Student Instructor (TA) — *Jan. 2018 – Dec. 2019*

- Led discussions and labs for 50 students in Devices & Systems and Computer Architecture, covering circuit design, control and signals theory, and computer architecture.
- Developed application-based labs for Optimization and Signals courses on stock price tracking, communication systems, and image compression.

Apple — Cupertino, CA
Product Security Software Engineering Intern — *May 2017 – Aug. 2017*

- Built a CLI for querying a distributed graph database to surface relationships between cybersecurity entities; implemented a honeypot server automation system.

Education

University of California, Berkeley — Berkeley, CA
M.S. / B.S., Electrical Engineering & Computer Science — *May 2021 / May 2020*

Technical Skills

Languages: Python, C++, CUDA, SQL, R, Matlab, Swift
ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers
Edge & Deployment: Quantization Aware Training (QAT), TensorRT, Qualcomm SNPE/QNN, ONNX
MLOps & Infrastructure: Docker, Weights & Biases, MLflow, distributed training (multi-GPU/cloud)
Libraries: OpenCV, scikit-learn, NumPy, SciPy, Pandas
Domains: Deep Learning, Computer Vision, Machine Learning, Computational Photography, Image Processing, Data Science, Optimization, Signal Processing, Edge AI

Projects

Computational Photography Projects

Colorizing Prokudin-Gorskii | Image Hybrids & Blending | Face Morphing | CNN Segmentation | Image Mosaicing | HDR & Seam Carving

RISC-V FPGA CPU & Audio Synthesizer [Report]

Built a 60 MHz 3-stage pipelined RISC-V CPU in Verilog on FPGA with data and control hazard handling, connected via memory-mapped I/O to a monophonic subtractive synthesizer capable of sinusoidal, sawtooth, square, and triangular waveforms.

Facial Recognition via Low-Dimensional Representations [Paper]

Improved dictionary-based facial recognition using Decimation, Haar Wavelet, PCA, and Convolutional Autoencoder representations.

Radio Modem & Image Compression [Report]

Transmitted an image across a room via a custom AFSK modem paired with a custom JPEG-inspired compression pipeline using color transformation, Haar wavelet, quantization, and LZW encoding.

Publications

S. Arora, V. Roeloffs, M. Lustig. "Untrained modified deep decoder for joint denoising parallel imaging reconstruction." *ISMRM Annual Meeting*, 2020.

Patents

"Digital Night Vision Device Using Artificial-Intelligence Based Image Signal Processing"
US App. No. 18/983,296, filed Dec. 2024 — Pub. US 2025/0265681 A1

"Image Color Enhancement Using Machine Learning in a Night Vision Device"
US App. No. 19/537,470, filed Feb. 2026 — Patent Pending

"Night Vision Device Using Recurrent State Warping for Motion Compensation"
US App. No. 19/656,887, filed Apr. 2026 — Patent Pending